CHRIST (Deemed to be University)
Department of Computer Science
Master of Computer Applications

Course: MCA412-2 - Research Methodology
LAB Exercise IV –
Dataset Identification and Description.

Date of assignment: 30-09-2026
Date of submission: 03-10-2026

Name : Aditya Kumar

Reg. No : 2647203
Class : 2 MCA – B

Domain :
Marine Ecological Informatics & Bio-Optical Remote Sensing

Lab 4

Lab Exercise IV:
Dataset Identification and Description.

1. Dataset Identification

- Dataset: ESA Ocean Colour Climate Change Initiative (OC-CCI) In-Situ Bio-Optical Database (Version 3)
- Repository & Link: PANGAEA (https://doi.org/10.1594/PANGAEA.941318)
- Publication: Earth System Science Data (Valente et al., 2022; https://doi.org/10.5194/essd-14-5737-2022)
- License: Creative Commons Attribution 4.0 International (CC BY 4.0) — the dataset is open-source for academic use as long as proper attribution is given.

2. Source and Provenance

- Source: Curated on PANGAEA through the European Space Agency (ESA) Ocean Colour Climate Change Initiative (OC-CCI) project.
- Acquisition: Extracted from 27 international marine archives including NOMAD, SeaBASS, AERONET-OC, MOBY, BOUSSOLE, and regional research cruises collected between 1997 and 2021.
- Atmospheric Correction: In-situ radiometric measurements were fully normalized to account for bidirectional reflection effects (sun at zenith, nadir viewing), matching satellite Level-2 products.

3. Data Type and Size

- Type: The dataset contains tabular text files (CSV format) relating to each other via a unique station key (idx).
- Size & Coverage: Contains 151,673 rows and 3,458 columns covering global oceans, with 68,641 remote-sensing reflectance stations and 82,543 chlorophyll observations, totaling approximately 4.2 GB uncompressed.

4. Purpose and Key Variables

- Purpose: Primarily used as a standardized global benchmark for developing, tuning, and validating satellite ocean-color algorithms and climate data records.
- Spectral Bands: Remote-sensing reflectance (Rrs) is available across 948 distinct wavelengths from 313 nm to 1022 nm, including standard bands at 412, 443, 486, 510, 551, and 671 nm.
- Key Measured Variables: Fluorometric and HPLC-derived chlorophyll-a (chla_fluor, chla_hplc), total suspended matter (tsm), and diffuse attenuation coefficient (Kd).
- Inherent Optical Properties: Spectral algal pigment absorption (aph), detrital plus CDOM absorption (adg), and particle backscattering (bbp).

5. Relevance to Research Problem

- Target Isolation: Contains explicit spectral Rrs observations at 510 nm, 531 nm, and 551 nm, directly enabling formulation of the Rrs(531)/Rrs(510) ratio for isolating Noctiluca scintillans cell densities.
- Bio-Optical Baseline: Provides paired absorption (aph, adg) and backscattering (bbp) values to identify departures from typical Case-1 diatom waters.
- Algorithm Validation: Allows training and evaluating multivariate regression models against ground-truth station observations in the Indian Ocean.

6. Limitations and Mitigation

- Regional Data Sparsity: In-situ stations are heavily clustered in North American and European coastal waters, with fewer points in the open Arabian Sea. This can be mitigated by sub-setting northern Indian Ocean cruise data and supplementing with Bio-Argo float profiles.
- Measurement Method Differences: Chlorophyll measurements combine both HPLC and fluorometric techniques. This is handled by using the provided flag_chl_method column and prioritizing HPLC data where available.
- High-Angle Optical Noise: Reflectance uncertainties increase under large observation geometries. This is resolved by applying standard quality filtering and screening out high solar zenith angle records.